

A Memory That Only Looks Like One: The Recurrent State of a Hybrid Vision-Language Model Relays the Picture Rather Than Storing It

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Abstract

A hybrid vision-language model carries two kinds of memory. Its attention layers keep a growing cache: every token the model reads, including every patch of an image, is stored as a key-value entry that later tokens can look up. Its recurrent layers keep a single state of fixed size, and every new token overwrites part of that state. Prior work proved that a fixed-size state must eventually lose information. It does not tell us *when* a given model forgets, *what* it keeps while forgetting, or *what breaks* downstream. We answer these three questions for Zamba2-VL-7B, a model in which all 81 decoder layers carry a Mamba2 recurrent mixer and only 13 also carry attention.

First, just after reading an image the recurrent state holds the gist, not the details. A linear probe reads *which* object was in the picture at 92.5% from the best single layer, against a 12.5% chance rate. The same probe can barely read *how many* objects there were or *where* they sat.

Second, how fast that memory fades can be computed from the weights alone. Mamba2 multiplies its state by $\exp(\Delta_t A)$ at every token, so retention after k tokens is $\exp(A \sum \Delta_t)$, and each head has a half-life of $\ln 2 / (|A| \bar{\Delta})$ tokens. The median predicted half-life is 6.6 tokens in the 7B model, and 2.9 and 2.6 in its two smaller siblings, with nothing fitted.

Third, and this is the central result: the memory a probe appears to find at longer distances is mostly not memory. Splitting heads by predicted half-life confounds decay with write amplitude, because the same gate sets both, so we hold the gate fixed and split by the decay weight $|A|$ alone. The ordering then comes out exactly as decay predicts, but the amounts are impossible. Heads with a 2.3-token half-life, whose own parameters keep 4×10^{-9} of any trace after sixty-four tokens, still decode the object at 26.6% there. Something must be re-supplying them. Deleting the image’s key-value entries from all 13 attention layers immediately after the image, without touching any recurrent state, collapses the apparent memory: the fast third of heads falls from 51.6% to 7.8% at thirty-two tokens, below chance, and at matched write amplitude the deletion costs fast heads 2.4 times more than slow heads beside the image. The recurrent channel is passing along what attention still holds; it is not storing the picture.

A direct causal test agrees. Exchanging all 81 recurrent states between two runs that saw different images changes nothing: 100 of 100 answers still follow the run’s own image, with every swap verified. Exchanging seven of the thirteen attention caches flips 94 of 100 answers to the other image. Finally, the practical consequence for cache pruning: the redundancy premise behind visual token and key-value pruning survives on this architecture down to a quarter of the entries, but the cost of pruning harder roughly doubles with distance. Removing every visual entry costs 34 accuracy points beside the image and 59 at thirty-two tokens.

1 What Was Known Before, and What Is New Here

Linear-attention and state-space models replace the growing key-value cache of a transformer with a state of fixed size [1, 2, 3, 4, 5, 6]. A fixed state cannot grow with the context. For this reason, most deployed systems keep a few attention layers alongside the recurrent ones, and these hybrid models are now common [7, 8, 9, 10, 11]. This paper asks a simple question about such a model: what is the recurrent half actually holding?

We want to be clear about what is already known. Prior work established that a fixed-size recurrent state must eventually lose information from context; that premise is not ours. Jelassi et

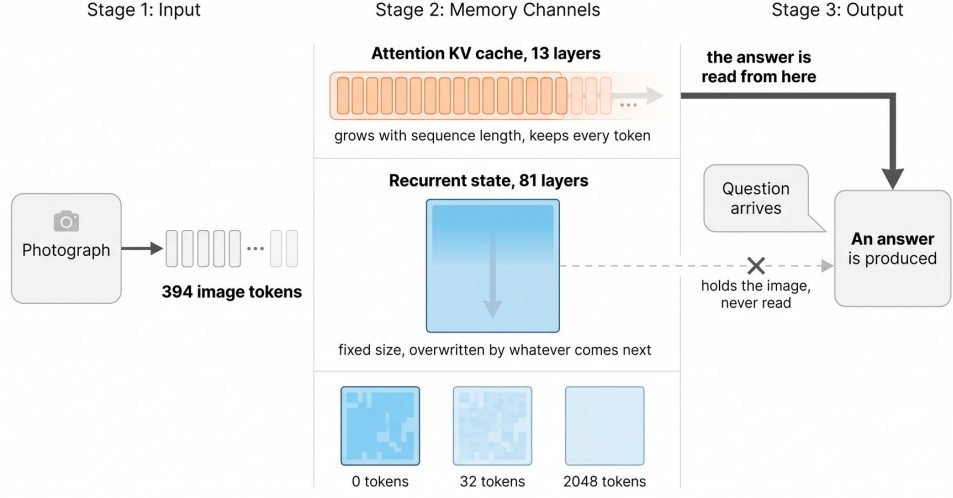


Figure 1: The two memory channels of a hybrid vision-language model and the question this paper asks of them.

al. [12] prove that transformers can copy exponentially long strings while state-space models are, in their words, “fundamentally limited by their fixed-size latent state”. They also show empirically that pretrained transformers “dramatically outperform state space models at copying and retrieving information from context”. Other results reach the same conclusion from other directions. State-space models are limited to TC^0 and cannot reliably track state [13]. In-context retrieval is the specific skill that separates recurrent models from transformers [14]. Most of the quality gap between efficient architectures and attention comes from recall [15, 16]. Our work builds on these results; it does not rediscover them.

Table 1 states what that literature establishes and what this paper adds.

Established by prior work	New here
A fixed-size state must lose information from context, with a capacity proof and behavioural evidence [12, 13, 14].	<i>When.</i> A retention horizon in tokens, computed from the model’s own A and Δ with nothing fitted, at three scales.
Recurrent models underperform at copying and retrieval as a class [15, 16].	<i>What survives.</i> The failure is content-selective: object identity is highly decodable at the same instant that count and position are near chance.
Architectures are compared to one another from the outside.	<i>Causal localisation inside one model</i> , and the finding that the recurrent channel relays rather than stores, which makes probing it at distance a measurement of the attention channel.
Visual tokens and key-value entries carry substantial redundancy that can be pruned [24, 25, 26].	<i>What underwrites that premise</i> , and that its safety margin shrinks with distance from the image on a hybrid.

Table 1: Delineation of prior work and contribution.

2 Significance

Why does this matter? Hybrid models are being adopted because a fixed-size state promises constant-memory inference. Anyone building or deploying one must then decide how much to trust the recurrent channel: can it be relied on to carry information across a long prompt? Today that is a judgement call. This paper turns it into a calculation. Given a checkpoint’s decay weight A and its

gate statistics, you can compute how many tokens the recurrent state retains what it wrote, before running any experiment. For Zamba2-VL-7B the answer is single digits. That is far shorter than one would guess from the size of the state, which is 458,752 numbers per layer.

There is also a concrete consequence for a popular optimization. A large body of work speeds up vision-language models by discarding visual tokens or evicting their key-value entries, on the stated premise that those entries are largely redundant [24, 25, 26]; a parallel line evicts key-value entries in language models on similar grounds [27, 28, 29]. These methods were developed and validated on dense transformers, where they work. We find that they mostly survive the move to this hybrid: a quarter of the visual entries is enough to hold accuracy at both distances we tested. What does not survive is the safety margin. The redundancy exists because the recurrent channel is relaying the picture, and it only relays for a few tokens. As a result, the cost of aggressive pruning roughly doubles between a prompt that asks about the image immediately and one that asks thirty-two tokens later. A pruning method tuned on short prompts will therefore look safer than it is. That is the kind of failure a standard benchmark is least likely to catch.

3 Model and Instruments

3.1 The model

Zamba2-VL-7B [35] is the vision-language member of the Zamba family [8], which pairs a Mamba2 backbone [5] with a shared attention module, and it takes its vision encoder from Qwen2.5-VL [31]. It is Apache-2.0 licensed and has 81 decoder layers. Every one carries a Mamba2 mixer, and 13 of them additionally carry a shared attention block, at layer indices 6, 11, 17, 23, 29, 35, 41, 47, 53, 59, 65, 71 and 77. The recurrent channel therefore spans all 81 layers and the attention channel spans 13. Each layer’s recurrent state is a tensor of shape [112, 64, 64], which is 458,752 numbers. One image occupies between 345 and 415 tokens depending on its aspect ratio.

3.2 Reading the state

The Mamba2 recurrence is

$$S_t = S_{t-1} \exp(\Delta_t A) + \Delta_t x_t B_t^\top, \quad y_t = S_t C_t, \quad (1)$$

so writing uses (x, B) , reading uses C , and retention of anything already present is governed by A and Δ alone. In words: at each token the state is first multiplied by $\exp(\Delta_t A)$, a number strictly between zero and one, which shrinks everything the state already holds. Then $\Delta_t x_t B_t^\top$ is added, which is the only operation that writes. The shrink factor is set by one stored weight A and one input-dependent gate Δ_t , so each head is a leaky accumulator whose leak rate can be read off the checkpoint. We verified each of these facts against the installed model source rather than assuming the canonical form. In particular $A = -\exp(A_{\log})$ is a per-head scalar, the gate is $\Delta = \text{softplus}(\Delta_{\text{raw}} + b)$ clamped below at 0.001 with no upper clamp, and the decay factor is $\exp(\Delta_t A)$ elementwise.

To measure what the state holds, we use a linear probe, the simplest possible reader: a classifier that computes a weighted sum of the state’s numbers and predicts the trial’s answer from that sum alone, trained on some trials and tested on held-out ones. If even this simple reader recovers the answer, the answer is present in an easily readable form. Probing this way is standard [17], and its hazards are well known. A probe can succeed because the classifier is powerful, not because the model uses the information. This is why probing papers check selectivity against control tasks [18] and read their results with care [19]. We therefore never rely on a probe alone. Every probe result is paired with an intervention on the model, in the tradition of causal mediation analysis [20] and activation patching [21, 22, 23], and we report what each intervention does to the model’s own behaviour, not only to the probe’s readout [23].

One practical detail: 458,752 numbers per layer is too wide to probe directly, so each layer’s state is first passed through a fixed Gaussian random projection down to 512 features. By the Johnson-Lindenstrauss property, such a projection preserves linear separability up to a controlled distortion, so a linear probe on the projected features is, to that tolerance, a linear probe on the state itself. The projection columns are drawn independently, so the first k columns are themselves a valid k -feature projection.

3.3 Splicing the channels

The splice runs the model twice, on two different images, and then exchanges one memory channel between the runs while leaving the other untouched. The run that keeps its own question is called the host; the run that donates its memory is called the donor. After the exchange, the host answers its question, and we record whose image the answer describes. Swapping the recurrent channel replaces all 81 states and convolution buffers; swapping the attention channel replaces the 13 key and value caches. The host’s cache is restored exactly afterwards, so one prefill supports many conditions.

4 Materials

Trials are built from COCO val2017 instance annotations [34], the corpus underlying most visual instruction tuning [30]. There are three question families: which object is present, how many of a named object are present, and where in the picture a named object is. Every question is eight-way forced choice, so the chance rate is exactly 12.5%. Trials are constructed in blocks of eight. All eight trials in a block share one identical candidate list, and each candidate is the correct answer exactly once within the block. This construction is what makes the pre-image control meaningful. Because the candidate set is fixed and each answer appears equally often, a classifier that has not seen the image has nothing to exploit: no candidate is more likely than any other, so it cannot beat chance.

Images are fetched at native geometry for the decay experiments. For the splice they are letterboxed onto a common square canvas. The reason is practical: two runs on images of different shape emit different numbers of vision tokens, and attention caches of different length cannot be exchanged. Letterboxing rather than square resizing matters. In a pilot, a plain square resize deformed the content badly enough to drop identity accuracy from 17 of 30 to 8 of 30.

5 Result 1: The State Records Gist, Not Particulars

We first probe the state at distance zero, the instant the image has just been read. The state is highly informative about which object is present, and close to uninformative about how many objects or where they are. Probing the full state across all 81 layers reads identity at 83.1% against a 12.5% chance rate ($p = 6 \times 10^{-92}$, $n = 160$), count at only 19.4% ($p = 0.009$), and position at only 17.5% ($p = 0.041$).

In fact these figures understate what the state carries. The full-state feature vector has $81 \times 512 = 41,472$ dimensions, far more than the number of trials, so the full-state probe overfits, and a probe on a single layer does substantially better. Layer 64 alone reads identity at 92.5%. The late third of the stack (layers 60 to 81) averages 85.2%, against 56.0% for the middle third. On the identical trials the full-state probe reads 71.7%. Two conclusions follow. The gist is stronger than the aggregate number suggests, and it is concentrated in the late layers.

The recurrent state is not a compressed copy of the image. It is a selective summary that keeps object identity and discards count and position, and that selection is visible at the instant of writing rather than emerging over distance.

6 Result 2: The Forgetting Horizon, Predicted From the Weights

6.1 The prediction

Retention of whatever the state held, after k further tokens, is

$$R(k) = \exp\left(A \sum_{t=1}^k \Delta_t\right), \quad \text{half-life} = \frac{\ln 2}{|A| \bar{\Delta}}. \quad (2)$$

A is a stored weight. $\bar{\Delta}$ is measured from the gate channels of the input projection over the tokens that follow the image, on the same prompts the decay experiment uses. Nothing here is fitted to a decay curve.

Across all $81 \times 112 = 9,072$ heads the predicted median half-life is 6.6 tokens, with an interquartile range of 2.8 to 17.8. 85.0% of heads fall below 32 tokens and 0.08% above 512. Measured over

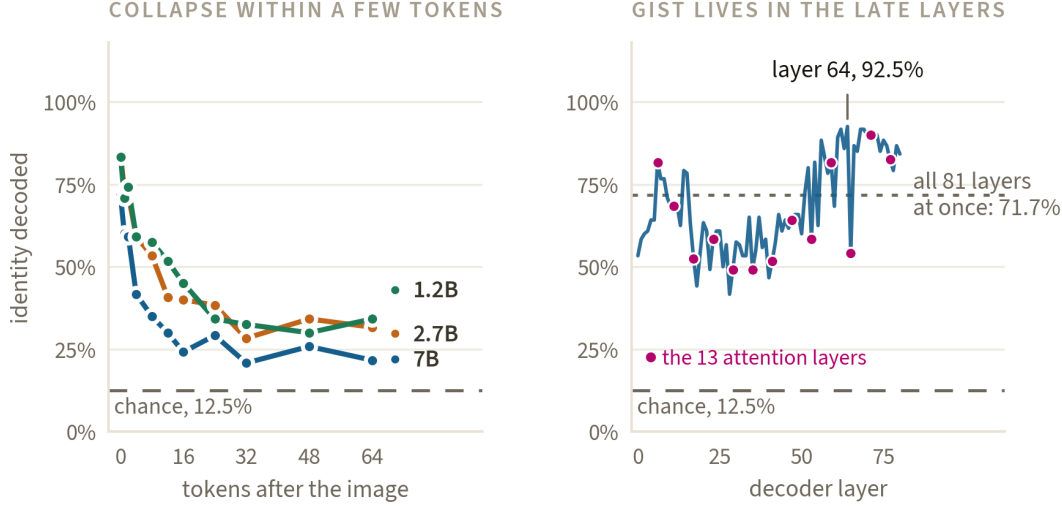


Figure 2: Left: identity decoded from the recurrent state against distance from the image, at three model scales, 120 trials per point. The pre-image control reads exactly 12.5% at every scale. **Right:** identity decoded from each layer’s state alone at the instant of writing, with red marking the 13 layers that also carry attention. A probe given all 81 layers at once does worse than the best single layer, because 41,472 features on 120 trials is heavily overparameterised.

eight different photographs the median moves only between 6.44 and 6.86 tokens, so the horizon is a property of the model rather than of the stimulus.

6.2 The measurement

Does the measured decay match this prediction? Our original collection measured only distances 0 and 32, and 32 tokens is already far past the region where the prediction changes quickly. Table 2 therefore shows a finer sweep, on 120 identity trials at each of eleven distances.

tokens	probe accuracy	p	behaviour	predicted retention (median)
0	71.7%	2×10^{-50}	94.2%	1.000
1	60.0%	2×10^{-34}	96.7%	0.901
2	59.2%	2×10^{-33}	97.5%	0.812
4	41.7%	1×10^{-15}	96.7%	0.659
8	35.0%	2×10^{-10}	97.5%	0.434
12	30.0%	3×10^{-7}	96.7%	0.286
16	24.2%	3×10^{-4}	97.5%	0.188
24	29.2%	1×10^{-6}	95.8%	0.082
32	20.8%	7×10^{-3}	95.0%	0.036
48	25.8%	6×10^{-5}	97.5%	0.007
64	21.7%	3×10^{-3}	98.3%	0.001

Table 2: Decodability of object identity against distance from the image, on 120 trials at every distance, with the retention predicted from the weights alongside. Chance is 12.5%. The pre-image control reads exactly 12.5%.

The measured signal halves at 4.0 tokens, which falls inside the predicted interquartile range of 2.8 to 17.8. Normalised to its value at distance zero, measured decodability correlates with predicted mean retention at $r = 0.973$ across the eleven distances, with no free parameters. Beyond 16 tokens the measured curve sits above the predicted median and between the median and the mean. The heavy tail of slow heads predicts exactly this: the residual signal at long distance is carried by the minority of heads with long half-lives.

Throughout this collapse the model’s own accuracy never moves. It answers correctly on 94 to 98% of trials at every distance, including those where the state has been emptied.

6.3 The within-model test, with write amplitude held fixed

A curve comparison alone cannot rule out coincidence. So we ran a sharper test. Every head has its own predicted half-life, so we can sort them: a *fast* head, with a half-life around two tokens, should empty almost immediately; a *slow* head should hold information longest. We probe the fast group and the slow group separately, using the same projection basis so the groups are comparable to each other and to the unstratified probe. If the state is genuinely retaining, decay makes a clear prediction: whatever signal survives at distance should sit in the slow heads.

Before giving the result, we must explain a confound, because it changed our own conclusions. The half-life is $\ln 2 / (|A| \bar{\Delta})$, and the gate $\bar{\Delta}$ appears in the write term as well, so the gate does double duty: it sets how fast a head forgets, and it also sets how strongly the head writes, by about a factor of seven between the fastest and slowest thirds. Why does writing harder matter? Because a head that writes with larger numbers leaves a stronger signal in its state, and a stronger signal is easier for a probe to read, whatever it contains and however old it is. Sorting by half-life therefore also, accidentally, sorts by signal strength. The naive split indeed produces a striking reversal, with the fast third decoding identity at 61.7% after sixty-four tokens while the slow third falls to 16.7%, the opposite of the decay ordering. That reversal has nothing to do with memory. It is the write strength.

The fix is to compare only heads that write equally hard. We keep the 3,628 heads whose mean gate falls in the middle forty percent of its distribution, and split those by the decay weight $|A|$ alone. This gives two strata of 907 heads each. Their predicted median half-lives still differ sevenfold, 2.3 against 16.4 tokens, but their write gains are now nearly equal (median $\bar{\Delta}$ of 0.074 against 0.066).

With the gate controlled, the reversal disappears (Table 3, left panel of Figure 3). The slow stratum decodes better at all seven distances measured, and it keeps a larger fraction of its distance-zero signal at sixty-four tokens, 0.28 against 0.20. The ordering that the decay parameters predict is real; amplitude had been masking it.

stratum	predicted $t_{1/2}$	$d=0$	$d=16$	$d=32$	$d=64$	signal kept at $d=64$
fast	2.3	81.8%	50.0%	35.9%	26.6%	0.20
slow	16.4	91.7%	61.5%	46.9%	34.4%	0.28

Table 3: Identity decodability by gain-matched stratum, $n = 192$ per cell, chance 12.5%. “Signal kept” is the above-chance excess at $d=64$ as a fraction of the excess at $d=0$. The slow stratum keeps more at every distance, which is the decay ordering; both strata keep far more than their own decay parameters permit.

But now look at the amounts, because they are the finding. A stratum with a predicted half-life of 2.3 tokens retains $2^{-64/2.3} \approx 4 \times 10^{-9}$ of anything it wrote sixty-four tokens ago. Yet it reads 26.6% there, twice the chance rate ($p = 1 \times 10^{-7}$, $n = 192$), and the measured half-life of its readout is 19.3 tokens, eight times its prediction. The slow stratum overshoots in the same direction: predicted 16.4 tokens, measured 26.3, with four times more signal surviving at $d=64$ than its parameters allow. In short, the strata are ordered exactly as decay predicts, and both hold far more signal than decay permits.

One caveat applies to any global split of this kind. The strata draw unequal numbers of heads from each layer, and depth matters a great deal for decoding, so the comparison between strata carries a depth component. The comparison of each stratum with its own prediction does not: whatever layers the fast stratum draws from, its own decay parameters bound what it can retain, and it reads far above that bound.

With write amplitude held fixed, the strata are ordered exactly as decay predicts, and both hold far more signal than decay permits. A head with a two-token half-life cannot be holding something written sixty-four tokens earlier: by then its state consists almost entirely of recent writes. If such heads still decode object identity at twice chance, the identity information in their state arrived recently. Something else in the model must be supplying it.

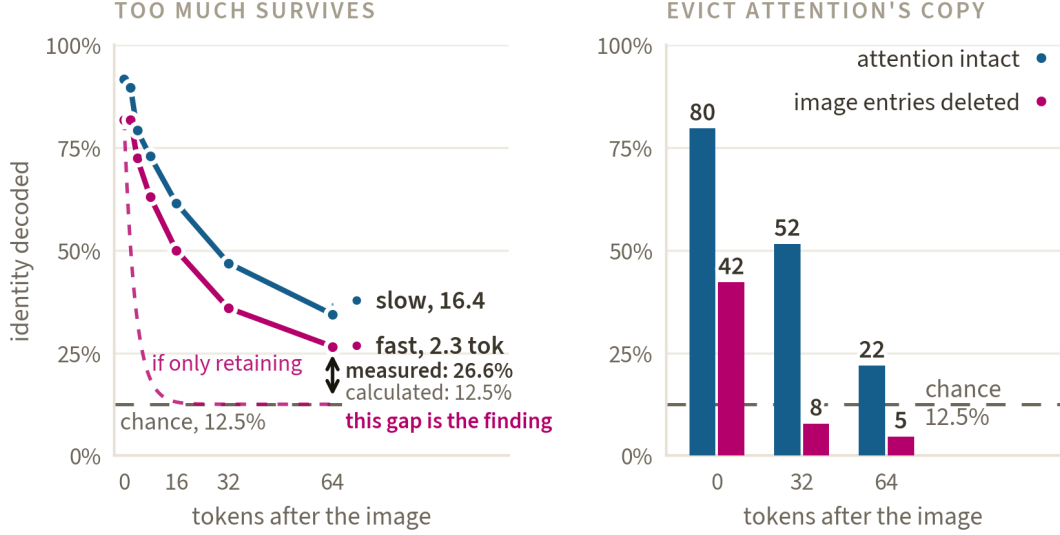


Figure 3: Left: how to read this panel. The two solid curves are measurements: identity decoded from the slow and fast strata at matched write gain, $n = 192$ per point. The dashed curve is not a measurement but a calculation: where the fast stratum would read if its signal came only from what its own 2.3-token half-life retains. The calculation reaches chance within ten tokens; the measurement ends at 26.6% at $d=64$. The slow-above-fast ordering is what decay predicts; the gap between the two fast curves is what decay cannot explain. **Right:** the same recurrent state probed with and without the attention channel’s copy of the image, fast tercile, $n = 64$ per bar. The eviction happens immediately after the image is read and never touches the recurrent state, which is identical in both runs at that instant.

7 Result 3: The Recurrent Channel Is a Relay, Not a Store

What could be supplying it? Only one component of the model still holds the image at distance: the attention channel, whose key-value entries do not decay. We tested this directly. Each trial was run twice, identically, except that in one run we deleted the image’s key-value entries from all 13 attention layers immediately after the image was read, *before a single filler token was processed*. The deletion does not touch the recurrent state. At that instant, the recurrent state is identical in both runs.

The two hypotheses now make opposite predictions. If the recurrent state is remembering the image on its own, deleting attention’s copy should not change what a probe reads from the state. If the state is being refreshed from attention, the deletion should collapse decodability at distance, and it should hurt the fast heads most, because they depend most on being refreshed.

stratum	distance	visual KV intact	visual KV evicted	change
fast	0	79.7%	42.2%	−37.5
fast	32	51.6%	7.8%	−43.7
fast	64	21.9%	4.7%	−17.2
middle	0	62.5%	43.8%	−18.8
slow	0	62.5%	54.7%	−7.8
all	0	59.4%	59.4%	0.0
all	32	6.2%	6.2%	0.0

Table 4: Identity decodability from the recurrent state with and without the attention channel’s copy of the image, $n = 64$ per cell. Chance is 12.5%. The fast heads’ apparent memory at 32 tokens is entirely dependent on the attention channel: removing it drops them from 51.6% to below chance.

There is a fair objection to the eviction experiment: deleting entries shortens every cache and changes the softmax normalisation, so perhaps we damaged the machinery rather than removed a memory. The control is to delete the same *number* of positions from the same caches, but delete

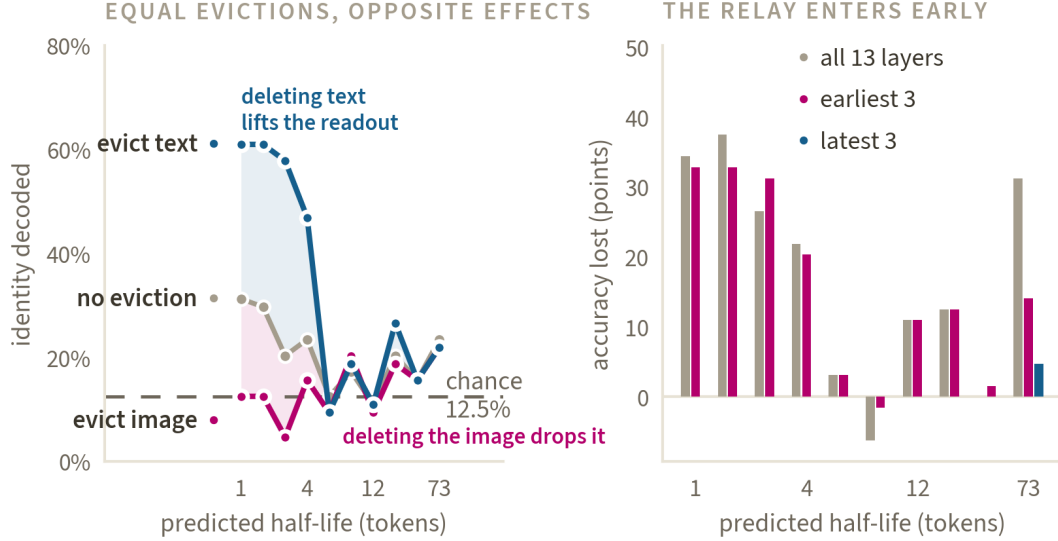


Figure 4: Left: read this panel vertically. At every x position, compare each curve with the no-eviction baseline: the text-eviction curve sits above it everywhere, the image-eviction curve below it. Both deletions remove the same number of positions from the same 13 caches, so damage alone cannot explain opposite signs. **Right:** accuracy lost when the visual entries are zeroed at all 13 attention layers, at only the earliest three, and at only the latest three. The earliest three reproduce most of the full effect; the latest three do nothing, so visual information enters early.

text instead of the image. The two versions do identical mechanical damage, so if damage were the explanation, both should push the reading down. Instead they move it in opposite directions (Figure 4, left): at 32 tokens, deleting the image costs the fastest decile 18.8 points while deleting an equal number of pre-image text positions *raises* it by 29.7; averaged over ten deciles the effects are -6.4 and $+12.5$ points. Damage cannot push a reading up.

7.1 The decay parameters predict how much is relayed

Result 2 found that the parameter-derived horizon does not predict the measured decodability curve. It does predict something else, and that something is exactly what the relay account says it should predict. Reason it through: a head that forgets quickly must depend more heavily on being refreshed, so deleting the source of the refresh should hurt its readout more. Ordering the strata by decay rate shows exactly this pattern. The size of the effect depends on how the strata are formed:

split	stratum	predicted $t_{1/2}$	intact at $d=0$	drop under eviction
tercile	fast	1.98	79.7%	37.5 points
tercile	slow	28.96	62.5%	7.8 points
gain-matched	fast	2.30	52.6%	22.4 points
gain-matched	slow	16.35	69.8%	9.4 points

Table 5: Dependence on the attention channel at $d=0$, against the decay rate implied by each stratum’s own A and gate. Terciles: $n = 64$ per cell; gain-matched: $n = 192$ per cell. The tercile contrast reaches $4.8\times$ but confounds decay with write amplitude; at matched amplitude the ratio is $2.4\times$. The two splits come from different collections, so accuracies are comparable within a split, not across splits.

Note that each row of this table compares intact against evicted *on the same heads*, so write amplitude cancels within a row. The amplitude confound affects only comparisons between strata. This is why the ratio worth quoting is the gain-matched one. At thirty-two tokens the gain-matched fast stratum reads only five points above chance even when intact, so eviction has little left to remove, and the contrast is uninformative there. The decay calculation therefore does carry predictive content,

but about the mechanism rather than about the decay curve: it identifies which heads are relaying and which are retaining. The effect is half the size the confounded split suggested, and it is measured at one distance on two strata. We flag both points in the limitations.

The headline contrast is large. At 32 tokens, the fast tercile reads 51.6% with the attention channel intact and 7.8% without it, which is below the 12.5% chance rate. In raw counts, thirty-three correct predictions out of sixty-four become five. Meanwhile the slow heads at distance zero, whose content was written while the image was still being read, are barely affected by eviction (62.5% to 54.7%). This is the control showing that the intervention does not simply break the model.

What a probe reads out of the recurrent state at distance is not what the state remembered. It is what the attention channel is still supplying, continuously re-supplied while that copy is present. Our evidence does not distinguish re-encoding at every token from a single re-supply shortly after the image, so we claim the dependence, not its period. The recurrent channel relays the picture; it does not store it.

7.2 What the probe can see, and why that matters here

There is one more objection to consider: perhaps the probe is simply insensitive to weak signals. Freshly re-encoded content plausibly sits at higher amplitude than content written long ago and decayed. A probe that finds strong signals and misses faint ones could then produce all of our observations even if the state never stopped storing the image.

This objection can be measured, and we measured it. We take the distance-zero features, where the image is present at full strength, and artificially weaken each trial’s own contribution while substituting an equal amount of unrelated variation from another trial, keeping total variance roughly constant: $x_i(\alpha) = \mu + \alpha(x_i - \mu) + \sqrt{1 - \alpha^2}(x_j - \mu)$. This transformation mimics what decay followed by overwriting does to a stored trace: the signal shrinks while unrelated content takes its place.

The result gives the probe a detection floor. The probe loses half its signal at $\alpha = 0.78$ and cannot be distinguished from chance by $\alpha \approx 0.25$. The best single-layer probe behaves the same way, halving at $\alpha = 0.75$. In plain terms, this instrument cannot see content below roughly a quarter of full amplitude.

Now put that floor next to the prediction. The weights say that after thirty-two tokens, a retained trace sits at 3.7% of full amplitude at the median head and 18% at the mean, both below the floor. So a genuinely retained trace at that distance *could not* produce a visible reading. Yet the intact probe reads the fast heads at 51.6% there. Whatever the probe is reading must therefore be high in amplitude, which means recently written. And deleting the attention channel’s copy removes it.

The probe’s insensitivity to faint signals is what makes the positive reading informative. A signal the instrument can see at thirty-two tokens is necessarily a fresh one, because a retained one would be four to six times below its detection floor.

The same measurement bounds what we may conclude. Whether faint retained content persists below the floor is not answerable with this instrument at any distance beyond a few tokens, so this paper does not claim the state stops storing the image. It claims that what a probe reads at distance is not retention, and that it depends causally on the attention channel.

This one measurement explains three earlier puzzles. First, it explains why the measured horizon exceeds the predicted one, and why the gap grows in smaller models: the probe was never measuring retention in the first place. Second, it explains why the tercile split made the fast heads look like the best retainers: they write hardest, so whatever the attention channel re-supplies lands in them at the highest amplitude, and holding the write gain fixed removes that apparent advantage (Section 6.3). Third, it supplies the mechanism behind the causal result, which the next section takes up.

8 Result 4: Causally, the Answer Follows the Attention Channel

The probe results say the attention channel supplies what the recurrent state appears to hold. The splice tests the same claim on the model’s own behaviour, with no probe involved. Recall the setup: two runs see different images, one memory channel is exchanged between them, and the host run then answers its question. We admit a pair only if the host and the donor each answered their own question correctly beforehand, so the ceiling is 100% by construction. Of 111 attempted pairs, 11

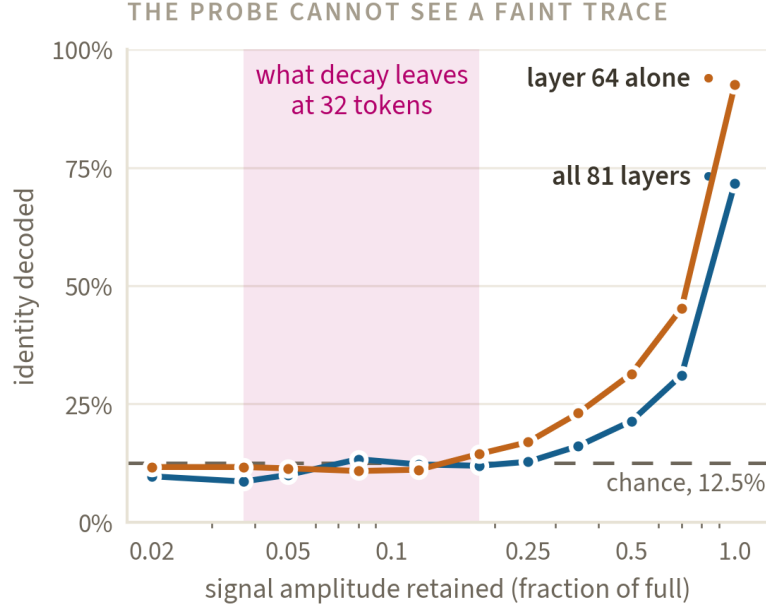


Figure 5: How faint a signal the probe can still recover. Each trial’s own contribution is attenuated while an equal amount of unrelated variation is substituted, holding total variance roughly constant. The shaded band is what the model’s decay parameters leave of a stored trace after 32 tokens, between the median and the mean over heads. Both probes are at chance throughout that band, so a visible reading at 32 tokens cannot be a retained one.

were rejected on that criterion, and none were rejected for mismatched cache length. Every recurrent swap was verified to have changed the state, 100 of 100.

condition	layers swapped	follows host	follows donor
no swap	0	100/100 [96, 100]	0/100 [0, 4]
recurrent, all 81 states	0	100/100 [96, 100]	0/100 [0, 4]
attention, 1 layer	1	100/100 [96, 100]	0/100 [0, 4]
attention, 3 layers spread	3	98/100 [93, 99]	0/100 [0, 4]
attention, 7 layers	7	4/100 [2, 10]	94/100 [88, 97]
attention, all 13	13	0/100 [0, 4]	98/100 [93, 99]
attention, earliest 3	3	94/100 [88, 97]	1/100 [0, 5]
attention, latest 3	3	100/100 [96, 100]	0/100 [0, 4]
both channels	13	0/100 [0, 4]	100/100 [96, 100]

Table 6: The splice at 100 pairs with a 100% ceiling. Brackets are 95% Wilson intervals in percent. Replacing the entire recurrent memory of all 81 layers changes nothing; replacing seven of the thirteen attention caches flips the answer to the other image.

Three things can be read from Table 6 and Figure 6. First, replacing the recurrent memory has no effect at all: exchanging all 81 recurrent states and convolution buffers leaves 100 of 100 answers following the host’s own image, with every swap verified to have taken effect. Second, the attention channel is causally sufficient to determine the answer: swapping seven of the thirteen caches moves 94 of 100 answers to the donor’s image. Third, the dependence on attention is graded and distributed. One swapped attention layer changes nothing, three spread across the stack move two answers, and seven move almost all of them. No small subset of layers is enough, so the picture is held across the attention layers rather than concentrated in a few. The earliest three layers have a small effect and the latest three have none, which matches visual information entering early and being consumed downstream.

Why does replacing the entire recurrent memory change nothing? Result 3 already answers this.

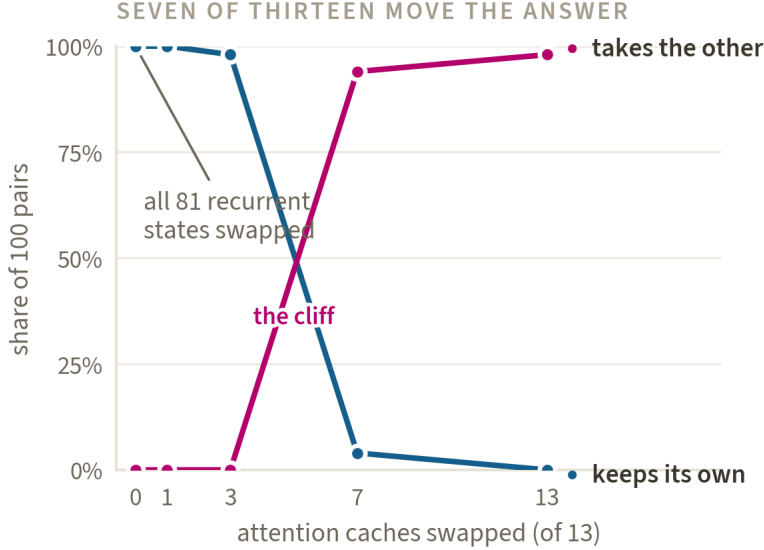


Figure 6: How many of the thirteen attention caches must be exchanged before the answer follows the other image, over 100 pairs in which both images were answered correctly beforehand. The leftmost point is the recurrent swap: all 81 states replaced, no attention cache touched, no effect.

The transplanted state is decayed away within a few tokens and rewritten from the attention channel, which the recurrent swap leaves untouched. The probe result and the splice result are the same fact seen from two sides.

In the earlier version of this work, the same splice gave a host baseline of 17 of 30. That figure was an artefact of our own instrument: the question was fed through the cache without the generation prompt the model expects, so the model often continued the prompt instead of answering. Correcting the prompt format lifts the baseline to 100% and leaves the direction of every conclusion unchanged.

9 Result 5: Scale, and Where the Prediction Fails

We repeated the core protocol at three model scales. The instruments read every tensor shape from the loaded model’s configuration rather than assuming the 7B geometry. This is a requirement, not a convenience: the 2.7B and 1.2B checkpoints use a single B/C group where the 7B uses two, and the 1.2B has a state dimension of 128 where the others have 64. Table 7 reports the results.

model	layers	attn	probe $d=0$	probe $d=64$	measured $t_{1/2}$	predicted $t_{1/2}$
Zamba2-VL-7B	81	13	71.7%	21.7%	4.0	6.64
Zamba2-VL-2.7B	54	9	84.2%	31.7%	9.6	2.91
Zamba2-VL-1.2B	38	6	83.3%	34.2%	14.3	2.58

Table 7: The same protocol at three scales, 120 identity trials per distance. The pre-image control reads exactly 12.5% at every scale. Half-lives are in tokens: measured is the distance at which above-chance decodability halves, predicted is the median over heads of $\ln 2 / (|A| \bar{\Delta})$.

Two conclusions follow. The second is a negative result, and we report it plainly.

First, the parameter-derived horizon is short at every scale. Median predicted half-lives are 6.64, 2.91 and 2.58 tokens, and the fraction of heads with a predicted half-life under 32 tokens is 85.0%, 97.9% and 95.6%. Whatever the recurrent state genuinely retains of its own past, it retains it for single-digit token counts in all three models.

Second, the predicted horizon does not predict the measured one. Across the three models the two quantities move in opposite directions: the prediction falls with scale while the measurement rises. On the 7B alone the agreement looked strong, with measured decodability tracking predicted retention at $r = 0.973$ across eleven distances. That agreement does not replicate at the other scales.

We therefore do not claim that the decay calculation explains the measured decay curve. Result 3 already showed why it cannot: the probe at distance is reading relayed content, not retained content, so its curve is not governed by the decay parameters.

10 Result 6: Visual Key-Value Eviction Removes the Only Copy, Eventually

Pruning methods discard visual key-value entries on the premise that they are redundant. Our results say the redundancy comes from the relay, and the relay is short-lived, so the premise should degrade with distance. We test this directly on the model’s own behaviour. We evict a fraction of the visual key-value entries from all 13 attention layers immediately after the image is read, let the filler and the question stream past the pruned cache, and measure the model’s accuracy. Retention ρ is the fraction of visual positions kept, sampled uniformly, and the question is asked d tokens after the image.

visual KV retained	$d=0$	$d=32$	$d=128$	$d=512$
100%	97.9%	95.8%	100.0%	97.9%
25%	91.7%	95.8%	95.8%	91.7%
5%	87.5%	77.1%	75.0%	64.6%
0%	64.6%	37.5%	31.2%	22.9%

Table 8: Behavioural accuracy under visual key-value eviction, against distance between the image and the question. $n = 48$ per cell, chance 12.5%, no failures. The unpruned row is flat, so distance alone costs nothing.

Table 9 reads the same data as cost, relative to the unpruned baseline at the same distance:

visual KV retained	$d=0$	$d=32$	$d=128$	$d=512$
25%	6.2	0.0	4.2	6.2
5%	10.4	18.8	25.0	33.3
0%	33.3	58.3	68.8	75.0

Table 9: Accuracy points lost to pruning, relative to the unpruned baseline at the same distance.

With a quarter of the entries retained, there is no distance effect at all: the cost is 6.2, 0.0, 4.2 and 6.2 points, flat within noise. At that level the redundancy premise holds on this architecture, and we found no evidence against it.

Below that level, distance starts to matter, and it matters a great deal. At a twentieth retained, the cost triples across the range, from 10.4 points beside the image to 33.3 points at 512 tokens. With every visual entry removed, it runs from 33.3 to 75.0. Compare the unpruned row, which is flat across the same distances. So the problem is not that long prompts are hard. The problem is that long prompts become hard *once the visual entries are gone*, because by then the recurrent channel has nothing left to relay from.

The cost of heavy pruning is not one number. Above a retention threshold, somewhere between a quarter and a twentieth on this model, pruning costs the same at every distance. Below it, the cost grows with how far the question sits from the image: a method validated at short range will understate its cost at long range by a factor of three at 5% retention.

The zero-retention row also gives the cleanest available measure of what the recurrent channel contributes on its own. With no visual entries anywhere in the attention caches, the model can only be answering from its recurrent states. Its accuracy falls from 64.6% beside the image to 22.9% at 512 tokens. That decline is the recurrent channel’s own contribution decaying, measured on behaviour with an instrument entirely independent of the probe, and it agrees with the picture the probe gives.

One discrepancy should be stated plainly. At zero retention the model still answers at 37.5% at thirty-two tokens, well above the 12.5% chance rate, while our linear probe reads the recurrent state at close to chance at that distance. So the model is using something in the recurrent channel that the probe does not recover. The probe is a lower bound on what the state carries, not a full measure of it. The sentence “the state is empty” is not supported by our evidence, and we do not claim it.

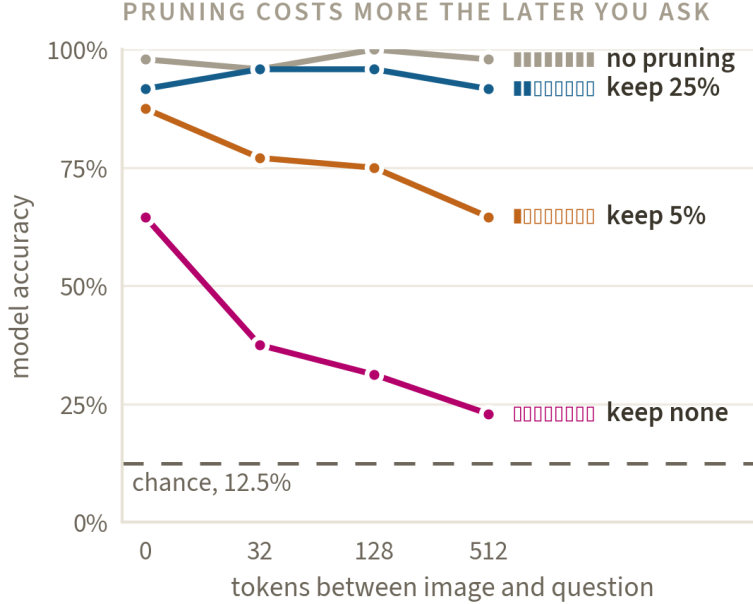


Figure 7: Model accuracy under visual key-value eviction, against how far the question sits from the image, $n = 48$ per point. The unpruned line is flat, so distance alone costs nothing. Keeping a quarter of the entries is also flat. Below that the lines fan out: the cost of pruning grows with distance only once the recurrent channel has stopped relaying the picture.

11 Limitations

We separate three kinds of limitation: alternative explanations our evidence does not exclude, limits on what the instruments can see, and limits on scope. The first kind is the one that should decide how much weight a reader puts on the central claim, so it comes first.

11.1 Alternative explanations we have not excluded

Eviction changes more than the image’s presence, though the matched control argues against this. Removing the visual key-value entries shortens every attention layer’s cache by roughly 400 positions and changes the softmax normalisation for every token that follows. The matched control (Section 7) deletes an equal number of *non-visual* positions from the same caches, and the two deletions move the readout in opposite directions, which pure damage cannot produce. A narrower version of the worry survives: the pre-image text we remove does not contribute to the residual stream the way the image does, so the comparison bounds the damage account rather than eliminating every version of it.

The relay result could be a sensitivity asymmetry. Our own data show the probe is a lower bound (below), and a lower bound need not be a uniform one. Freshly re-encoded content plausibly sits at higher amplitude in the state than content written many tokens earlier and decayed. If the probe recovers high-amplitude content easily and low-amplitude content poorly, then removing the re-encoded copy would collapse the readout even in a state that was still genuinely retaining something. That account predicts the same observations we report. Distinguishing it requires either a probe whose sensitivity is calibrated against signal amplitude, or a model with no attention channel to relay from, where probing the recurrent state must be measuring retention because there is nothing else it could be measuring.

The route is not traced. The eviction contrast shows that decodable identity at distance depends on the attention channel holding the image. It does not show the path. The natural account is that attention layers write image-derived content into the residual stream and downstream Mamba2 mixers re-encode it, but a layerwise eviction, removing the visual entries from only the early or only the late attention layers, would be needed to establish it and we have not run one.

11.2 What the instruments can and cannot see

The probe is a lower bound, and our own behavioural data prove it. With every visual key-value entry removed, the model answers at 37.5% at thirty-two tokens while the probe reads the recurrent state near chance at the same distance. Something usable survives there that a 512-dimensional random projection, read by a linear probe trained on 120 examples, does not recover. Every statement in this paper about what the state contains should be read as a statement about what is linearly decodable under that instrument. The paper does not establish that the state is empty of the image at any distance, and readers should not take the phrase “the state empties” from the earlier version of this work as supported.

Probe accuracy depends on the feature-to-trial ratio. Count read at chance with 128 projection features and reached 19.4% at 512. In the other direction, the full 41,472-dimensional state probe reads 71.7% where a single layer’s 512 features read 92.5% on identical trials. Absolute accuracies are therefore not comparable across collections of different size or width, and we report them only within a collection. The decay curve in Table 2 is internally consistent because the same 120 trials appear at every distance.

One projection basis, one seed. Every probe result uses a single fixed Gaussian projection. We have not repeated the analysis across projection seeds, so the variance attributable to the choice of basis is unmeasured.

The stratum dose-response is confirmed at matched amplitude only at distance zero. On the tercile split eviction sensitivity is ordered exactly by predicted decay rate (37.5, 18.8 and 7.8 points, $r = 0.991$), but every split by half-life is also a split by write amplitude. With the gate held fixed the ratio is $2.4\times$ at distance zero over two strata at $n = 192$, and unmeasurable at thirty-two tokens, where the gain-matched fast stratum reads too close to chance intact for eviction to have anything to remove. Two strata at one distance in one model is a direction, not a curve.

11.3 Scope

One model family, and a second family we tried and failed on. Results come from three Zamba2-VL checkpoints, which is one vendor, one architecture family and one training recipe. This is the principal limit on the paper’s generality and we do not argue around it.

We attempted to extend the relay result to a second family and report the attempt because its failure is informative. Granite 4.0 h-tiny [36] is a Mamba2 hybrid from a different lineage, with 36 recurrent layers to 4 attention layers against Zamba2-VL-7B’s 81 to 13. Its decay parameters read out cleanly and give a median predicted half-life of 8.7 tokens, so the horizon calculation of Section 5 transfers to a second family. The relay contrast does not, for an instrumental reason: on a code-word recall task the model answers correctly on 100%, 96.9% and 100% of trials at 0, 32 and 64 tokens, and yet a probe on its recurrent state reads at chance at distance zero. Scaling to 256 trials did not change this (13.7% against a 12.5% chance rate, $p = 0.31$), and neither did probing single layers, where the best of 36 layers reached 19.9%.

We suspected the item was too small to write a detectable trace, a one-token code word against a 345-token image, and tested that by repeating the carrier sixteen times so the item occupied 192 tokens. It did not help: 12.0%, 17.2% and 10.9% across the three strata. Repetition lengthens an item without diversifying what it writes, so the state ends up holding little more than the final repeat, which is not the manipulation we intended. We do not know why the instrument fails on this model; we only know that it does. No relay claim can be made beyond the Zamba2-VL family on this evidence. A vision-language hybrid from a second family, where the item is a real image, remains the right test and we have not run it. The sharper version of that test is a vision-language model with no attention layers at all, such as Cobra [32] or VL-Mamba [33], where probing the recurrent state must be measuring retention because there is nothing else it could be measuring. Both are Mamba1 models and would need the read operator ported before any number from them could be trusted.

One task format, one dataset, one modality. Eight-way forced choice on COCO buys an exact chance rate and a clean pre-image control, and costs generality to free-form answering, to other visual properties, and to whether any of this holds for text held in the same states.

The horizon is measured on one filler distribution. $\bar{\Delta}$ is input dependent. We measured it on natural prose and found the median half-life stable across images, from 6.44 to 6.86 tokens, but a

filler with very different gate statistics would give a different horizon, which the same calculation would predict.

An instrument artefact we corrected, and what it implies. The splice reported in the earlier version of this work fed the question through the cache without the generation prompt the model expects. The model frequently continued the prompt rather than answering it, naming every candidate and scoring as ambiguous, which is why the host baseline read 17 of 30. Under the corrected format the baseline is 100/100. The direction of every conclusion is unchanged, but the earlier absolute rates were a property of our instrument rather than of the model. We report both so the difference is auditable, and we note the general lesson: a null result measured through a degraded instrument is worth less than its sample size suggests.

12 Conclusion

The recurrent state of this hybrid vision-language model is not a memory of the image in any useful sense. It is a selective summary of the picture’s gist, and it holds that summary for only about four tokens. That lifetime is not an empirical curiosity: it can be computed from the model’s own decay parameters before any measurement is taken. What the model actually answers from lives in the thin attention channel, and the recurrent state relays that content rather than storing its own copy. The practical consequence is direct: the visual key-value entries a hybrid carries are redundant only for as long as the recurrent state is still relaying the picture, and that is not long.

References

- [1] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. *ICML*, 2020.
- [2] A. Gu, K. Goel, and C. Ré. Efficiently Modeling Long Sequences with Structured State Spaces. *ICLR*, 2022.
- [3] B. Peng, E. Alcaide, Q. Anthony, A. Albalak, S. Arcadinho, S. Biderman, et al. RWKV: Reinventing RNNs for the Transformer Era. *arXiv:2305.13048*, 2023.
- [4] A. Gu and T. Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *arXiv:2312.00752*, 2023.
- [5] T. Dao and A. Gu. Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality. *ICML*, 2024.
- [6] M. Beck, K. Pöppel, M. Spanring, A. Auer, O. Prudnikova, M. Kopp, G. Klambauer, J. Brandstetter, and S. Hochreiter. xLSTM: Extended Long Short-Term Memory. *arXiv:2405.04517*, 2024.
- [7] O. Lieber, B. Lenz, H. Bata, G. Cohen, J. Osin, I. Dalmedigos, et al. Jamba: A Hybrid Transformer-Mamba Language Model. *arXiv:2403.19887*, 2024.
- [8] P. Glorioso, Q. Anthony, Y. Tokpanov, J. Whittington, J. Pilault, A. Ibrahim, and B. Millidge. Zamba: A Compact 7B SSM Hybrid Model. *arXiv:2405.16712*, 2024.
- [9] S. De, S. L. Smith, A. Fernando, A. Botev, G. Cristian-Muraru, A. Gu, et al. Griffin: Mixing Gated Linear Recurrences with Local Attention for Efficient Language Models. *arXiv:2402.19427*, 2024.
- [10] L. Ren, Y. Liu, Y. Lu, Y. Shen, C. Liang, and W. Chen. Samba: Simple Hybrid State Space Models for Efficient Unlimited Context Language Modeling. *ICLR*, 2025.
- [11] R. Waleffe, W. Byeon, D. Riach, B. Norick, V. Korthikanti, T. Dao, A. Gu, et al. An Empirical Study of Mamba-based Language Models. *arXiv:2406.07887*, 2024.
- [12] S. Jelassi, D. Brandfonbrener, S. M. Kakade, and E. Malach. Repeat After Me: Transformers are Better than State Space Models at Copying. *arXiv:2402.01032*, 2024.
- [13] W. Merrill, J. Petty, and A. Sabharwal. The Illusion of State in State-Space Models. *ICML*, 2024.
- [14] K. Wen, X. Dang, and K. Lyu. RNNs are not Transformers (Yet): The Key Bottleneck on In-context Retrieval. *arXiv:2402.18510*, 2024.
- [15] S. Arora, S. Eyuboglu, A. Timalsina, I. Johnson, M. Poli, J. Zou, A. Rudra, and C. Ré. Zoology: Measuring and Improving Recall in Efficient Language Models. *arXiv:2312.04927*, 2023.

- [16] S. Arora, S. Eyuboglu, M. Zhang, A. Timalsina, S. Alberti, D. Zinsley, J. Zou, A. Rudra, and C. Ré. Simple Linear Attention Language Models Balance the Recall-Throughput Tradeoff. *arXiv:2402.18668*, 2024.
- [17] G. Alain and Y. Bengio. Understanding Intermediate Layers Using Linear Classifier Probes. *arXiv:1610.01644*, 2016.
- [18] J. Hewitt and P. Liang. Designing and Interpreting Probes with Control Tasks. *EMNLP*, 2019.
- [19] Y. Belinkov. Probing Classifiers: Promises, Shortcomings, and Advances. *Computational Linguistics*, 2022.
- [20] J. Vig, S. Gehrmann, Y. Belinkov, S. Qian, D. Nevo, S. Sakenis, J. Huang, Y. Singer, and S. Shieber. Causal Mediation Analysis for Interpreting Neural NLP: The Case of Gender Bias. *arXiv:2004.12265*, 2020.
- [21] K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and Editing Factual Associations in GPT. *NeurIPS*, 2022.
- [22] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small. *arXiv:2211.00593*, 2022.
- [23] F. Zhang and N. Nanda. Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. *ICLR*, 2024.
- [24] L. Chen, H. Zhao, T. Liu, S. Bai, J. Lin, C. Zhou, and B. Chang. An Image is Worth 1/2 Tokens After Layer 2: Plug-and-Play Inference Acceleration for Large Vision-Language Models. *ECCV*, 2024.
- [25] S. Yang, Y. Chen, Z. Tian, C. Wang, J. Li, B. Yu, and J. Jia. VisionZip: Longer is Better but Not Necessary in Vision Language Models. *arXiv:2412.04467*, 2024.
- [26] Y. Shang, M. Cai, B. Xu, Y. J. Lee, and Y. Yan. LLaVA-PruMerge: Adaptive Token Reduction for Efficient Large Multimodal Models. *ICCV*, 2025.
- [27] Z. Zhang, Y. Sheng, T. Zhou, T. Chen, L. Zheng, R. Cai, Z. Song, Y. Tian, C. Ré, C. Barrett, Z. Wang, and B. Chen. H₂O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. *arXiv:2306.14048*, 2023.
- [28] Y. Li, Y. Huang, B. Yang, B. Venkitesh, A. Locatelli, H. Ye, T. Cai, P. Lewis, and D. Chen. SnapKV: LLM Knows What You are Looking for Before Generation. *arXiv:2404.14469*, 2024.
- [29] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. Efficient Streaming Language Models with Attention Sinks. *ICLR*, 2024.
- [30] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual Instruction Tuning. *NeurIPS*, 2023.
- [31] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, et al. Qwen2.5-VL Technical Report. *arXiv:2502.13923*, 2025.
- [32] H. Zhao, M. Zhang, W. Zhao, P. Ding, S. Huang, and D. Wang. Cobra: Extending Mamba to Multi-Modal Large Language Model for Efficient Inference. *AAAI*, 2025.
- [33] Y. Qiao, Z. Yu, L. Guo, S. Chen, Z. Zhao, M. Sun, Q. Wu, and J. Liu. VL-Mamba: Exploring State Space Models for Multimodal Learning. *arXiv:2403.13600*, 2024.
- [34] T.-Y. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C. L. Zitnick, and P. Dollár. Microsoft COCO: Common Objects in Context. *ECCV*, 2014. *arXiv:1405.0312*.
- [35] Zephyra. Zamba2-VL-7B model card. <https://huggingface.co/Zyphra/Zamba2-VL-7B>, 2026. Apache-2.0.
- [36] IBM Granite Team. Granite 4.0 h-tiny model card. <https://huggingface.co/ibm-granite/granite-4.0-h-tiny>, 2025.